VDA10.3

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Exercise 3 (Direct Volume Rendering in ParaView, 10 Points)

a) Open ParaView and load the head.vti dataset from the previous sheet. In the “Properties” tab, hit “Apply” to actually make it available. Now, change the representation to “Volume” and confirm your choice. At this point, you should see a result as in Figure 4 left. Now, modify the transfer function and rotate the volume so that you obtain a result similar to Figure 4 right. Please submit a screenshot of your result, including the transfer function editor. (3P)

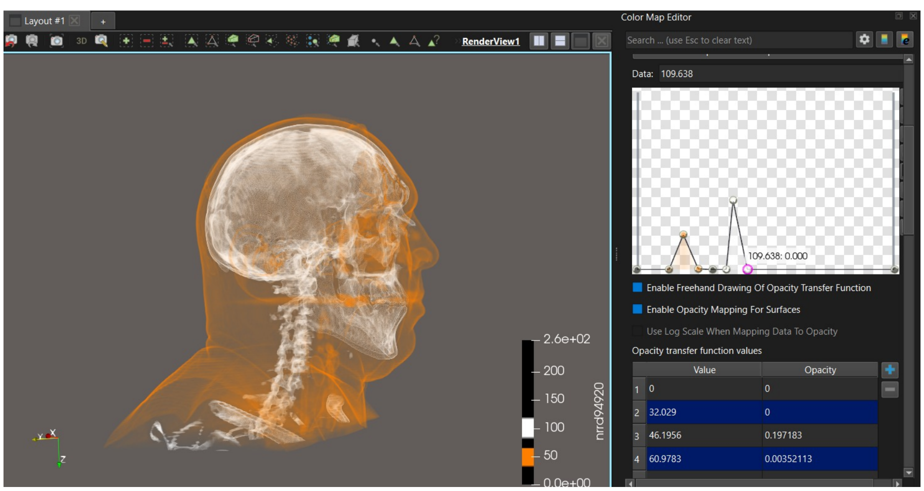

Figure 1: Screenshot of paraview with the visualization and the transfer function on the right

b) In Figure 5, a transfer function has been chosen that assigns full opacity, but only in a very narrow value range (close to a delta peak). What might be the intended effect of such a transfer function? Why does it not lead to cleaner image? Which method do you remember from the lecture that could improve the rendering in such a case? (3P)

The intended effect is probably the creation of an isosurface, as only the points with a particular value should be showing on the graph. The image is not clear due to the quantization of the measurements and the discretization of the domain.
A proposed solution given in the lectures is the use of the Levoy transfer function, which would take into account the magnitude of the gradient to set the opacity at each point.

c) Figure 6 left shows the result of using a step function as a transfer function, which visualizes the silhouette of a surface. Provide a screenshot that additionally reveals the surface shape, as in the middle image. What was missing in the first case, and had to be added for the second one? (2P)

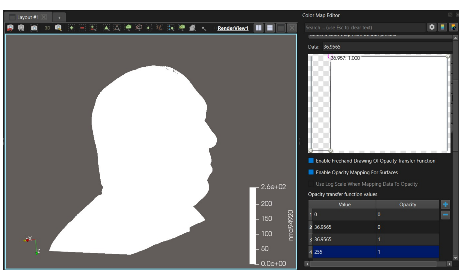

Figure 2: Volume rendering with a step function

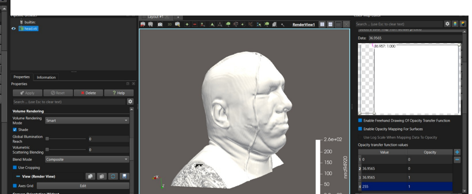

Figure 3: The same rendering with shading activated

We had to activate shading for the surface shape to become visible.

d) While rotating the volume, you will notice artifacts such as the ones highlighted by the red arrow in Figure 6 right. Explain why they vanish automatically once you release the mouse. (2P)

When rotating the volume, despite of the geometry apparently having the same level of detail, a series of interpolation and shading artifacts occur. Thus, I suspect that the programme skips some proccesing steps while moving the camera to keep the data frame constant. If the plotter used a post-classification algorithm, it wouldn't need to be recomputed when changing the view. Therefore the best explanation I can find is that the default settings use some kind of pre-classification algorithm to improve the visual appeal, and that this feature is turned off when moving the camera because it would be too expensive to compute every frame.